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Gauge invariance in the string action

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Introduction and conclusions

String theories represent one of the most current attempts to unify the four fundamental interactions in a single system [1]: the electromagnetic, weak and strong interactions, formalised by quantum field theories, together with gravity, which we describe through general relativity, a classical field theory. The first goal, still unresolved, which these theories try at least in part to clarify, is to understand how to build a theory of *Quantum gravity*, able to explain the quantum structure of spacetime, which could then be joined to the other *Quantum field theories* (QFT).
String theories begin by describing the interactions through fundamental 1D objects, the strings, whose low-energy vibration modes reproduce the behaviour of the particles in the traditional QFTs used for the standard model. In particular, from the point of view of their mathematical structure they turn out to be QFTs in a 2D space, the *world-sheet* of the string.
A property shared by these theories is that they are gauge theories and that, besides invariances common to the known field theories, they also exhibit others specific to 2D QFTs, such as Weyl conformal invariance. Today the study of these invariances, in particular following the AdS/CFT correspondence conjecture [2], enjoys great success and interest across the various lines of research.
The aim of this thesis is to introduce the reader to some of the elementary features of string theories, by studying the closed *Polyakov* string in flat spacetime.
In the first chapter we will introduce the classical Nambu-Goto and Polyakov strings and their formulations in the Lagrangian and Hamiltonian formalisms, and we will identify the symmetries and gauge invariances of the system.
In the second chapter we will discuss what it means to have gauge invariances and to decide to eliminate them from a Hamiltonian and symplectic point of view, introducing some properties of constrained Hamiltonian systems; we will then put what we discussed into practice on our system by applying the *uniform light-cone gauge* [3], and using the constraint conditions to compute the metric perturbatively.
Finally, in the third chapter we will first study the reduced system through a perturbative approach, following what was done in [3]: we will compute, in the decompactification limit, the Hamiltonian up to quartic order in the fields, the unperturbed solutions of the motion, the dispersion law for the free string and the tree-level Scattering matrix (generalising the result to the case of relativistic effective actions). Then, recalling results from integrability theory, we will discuss the computation of the spectrum of the Hamiltonian in finite volume, with a mention of the corrections and of the neglected contributions. This result will show that, although the theory seems to depend on a gauge parameter, the physical observables, in particular the spectrum, are in fact independent of the gauge choice, as we expect it should be.
Although the treatment is elementary and adopts a simple and well-known model for the analysis, it is enough to modify the ingredients only slightly to turn the questions analysed into topics of current research interest. For example, in superstring theory the action used is made of a fermionic term (itself subject to gauge-invariance issues) and of a bosonic one, described by an action analogous to Polyakov's. These superstrings can be placed in curved spacetimes, usually compatible with the application of the AdS/CFT correspondence, as done for $AdS_5 \times T^5$ in [3] or for $AdS_3 \times S^3 \times T^4$ in [11] or in [4]. In these cases the computation of the Scattering matrix and of the Hamiltonian spectrum requires bootstrap techniques that exploit properties related to the integrability of these models, and in several cases it is still an open problem.
Finally, the Nambu-Goto string turns out to be non-trivial enough to be used in the study of theories of *Quantum gravity* [5], and it allows one to analyse within the same theory phenomena such as

Quantum Black Holes and the Heisenberg uncertainty principle for strings.

The string in flat spacetime

1.1 Action of the relativistic particle

Before describing the properties of the string action, it may be useful to review the same problem in the case where one wants to describe a classical relativistic particle.
Consider a D-dimensional spacetime $\mathbb{R}^{1,D-1}$, whose metric follows the convention:

$$\eta_{\mu\nu} = \text{diag}(-1, +1, .., +1).$$

Requiring an action that is invariant under Lorentz transformations (i.e. a scalar) and compatible with the non-relativistic free-action limit, we obtain[1]:

$$S = -m \int_{-\infty}^{+\infty} \mathrm{d}t \, \sqrt{1 - \dot{X}^i \dot{X}^j \delta_{ij}}. \tag{1.1}$$

Here m is the mass of the particle and $X^\mu = (X^0, X^i)$ the four-position, with $i, j = 1, .., D-1$ [2].
An equivalent way to write the action so that it is manifestly Lorentz invariant consists in making explicit its dependence on the spacetime interval s, defined by the integral of the square root of $ds^2 = dX^\mu dX^\nu \eta_{\mu\nu}$, as:

$$S = -m \int_{-\infty}^{+\infty} \mathrm{d}s. \tag{1.2}$$

Moreover, if we want the theory to be invariant under changes of coordinates (as happens for example in general relativity), we can introduce a generic time-like parameter[3] τ, which represents the coordinate along the trajectory of the particle. After this addition, the action (1.1) can be rewritten in the more general form:

$$S = -m \int_{-\infty}^{+\infty} \mathrm{d}\tau \, \sqrt{-\dot{X}^\mu \dot{X}^\nu \eta_{\mu\nu}}. \tag{1.3}$$

Here $\dot{X}^\mu = dX^\mu/d\tau$. At this point it might seem that the theory has one more degree of freedom, since the fields to be determined in the action are now D and no longer $D-1$. In reality, inserting the parameter in (1.3) only makes implicit the dependence between X^i and X^0 in (1.1). This relation can be recovered by writing the conjugate momentum P^μ:

$$P_\mu := \frac{\partial L}{\partial \dot{X}^\mu} = m \frac{\dot{X}^\mu}{\sqrt{-\dot{X}^\nu \dot{X}_\nu}}, \tag{1.4}$$

and computing its square:

$$P^2 = P^\mu P_\mu = -m^2, \qquad P^2 + m^2 = 0. \tag{1.5}$$

[1] From now on we will use the system of natural units, in which $c = \hbar = 1$. In this notation it will be equivalent to write $X^0 = t$.

[2] From now on we will adopt Einstein's repeated-index notation, where the metric used to contract will be, unless stated otherwise, δ_{ij} for Latin indices and $\eta_{\mu\nu}$ for Greek indices.

[3] A parameter τ is called time-like if $g_{\mu\nu} \partial_\tau X^\mu \partial_\tau X^\nu < 0$ for every point X^μ parametrised by the curve.

One immediately sees that the momenta are not all independent but are related by an algebraic equation, which in this case is the dispersion law of the free particle.
Once the equations of motion (EOM) of the relativistic particle are known, we want to find a new action containing an additional field $e(\tau)$, such that the solutions for the four-positions X^μ remain unchanged and the Lorentz and reparametrisation symmetries are preserved. This is possible, for example, by writing:

$$S = \frac{1}{2}\int_{-\infty}^{+\infty} \mathrm{d}\tau \left(e^{-1}\dot{X}^2 - em^2\right). \tag{1.6}$$

From the EOM one obtains:

$$\partial_\tau(e^{-1}\dot{X}^\mu) = 0, \qquad \dot{X}^2 + e^2m^2 = 0. \tag{1.7}$$

Substituting e in the first EOM with what is obtained from the second one, one finds that the motion of a particle is equivalent to the one described by (1.1). It is possible to interpret e as: $e = \sqrt{-g_{\tau\tau}}$, where g is the metric induced by $\eta_{\mu\nu}$ on the trajectory of the particle (corresponding to a sub-Riemannian manifold of dimension 1). This is confirmed by the fact that reparametrisation invariance remains valid in the theory if e is taken to transform according to the interpretation defined above:

$$\tau \to \tau - \eta(\tau), \qquad \delta e = \frac{d}{d\tau}(\eta(\tau)e), \qquad \delta X^\mu = \frac{dX^\mu}{d\tau}\eta(\tau). \tag{1.8}$$

In light of what has been said, it is possible to give a further general interpretation of the theory just constructed: one can regard the theory described by (1.6) as a theory of gravity (that is, one for which the variational principle returns a metric) on a 1d manifold parametrised by τ, called *world-line*, on which D scalar fields X^μ live.

1.2 String action

What is a relativistic string? If a particle is represented as a point whose motion is described by a curve in spacetime, a string will be equivalent to a deformable circle[4], whose motion in spacetime describes a surface homeomorphic to a cylinder. This surface, determined by the EOM, is called the *world-sheet*. To describe a string through an action a new space-like parameter $\sigma \in [-R/2, R/2]$ is needed. The action can then be written as:

$$S = \int_{-\infty}^{+\infty} \mathrm{d}\tau \int_{-R/2}^{+R/2} \mathrm{d}\sigma\, \mathcal{L}(\tau,\sigma) = \int \mathrm{d}^2\sigma\, \mathcal{L}(\tau,\sigma). \tag{1.9}$$

Here $\mathcal{L}(\tau,\sigma)$ is the Lagrangian density, while in the second equality the convention $\sigma = (\sigma,\tau)$ is used.

1.2.1 Nambu-Goto string

The simplest action of the type (1.9) is obtained, in analogy with (1.2), by considering the invariant surface element, and it is called the *Nambu-Goto string*:

$$S = -T\int \mathrm{d}^2\sigma\sqrt{-\gamma}. \tag{1.10}$$

Here $\gamma = \det(\gamma_{\alpha\beta}) = \frac{\partial X^\mu}{\partial\sigma^\alpha}\frac{\partial X^\nu}{\partial\sigma^\beta}\eta_{\mu\nu}$ is the metric induced on the world-sheet by spacetime and T a constant which, in analogy with the mass for the particle, quantifies the tension of the string. In this case invariance under changes of coordinates can be seen by recognising that the integrand is an

[4] In general, strings are divided into closed strings if homeomorphic to $\mathbb{S}^1$ and open strings if homeomorphic to the interval $[0,1]$.

invariant volume form of the world-sheet, or by writing the determinant explicitly as a function of the fields:

$$S = -T \int \mathrm{d}^2\sigma \sqrt{-\dot{X}^2 \acute{X}^2 + (\dot{X}^\mu \acute{X}_\mu)^2}. \tag{1.11}$$

Here $\dot{X}^\mu = \partial_\tau X^\mu$, $\acute{X}^\mu = \partial_\sigma X^\mu$, $\dot{X}^2 = \dot{X}^\mu \dot{X}_\mu$ and $\acute{X}^2 = \acute{X}^\mu \acute{X}_\mu$. From the variational method, using the identity $\delta\sqrt{-\gamma} = \frac{1}{2}\sqrt{-\gamma}\gamma^{\alpha\beta}\delta\gamma_{\alpha\beta}$, one obtains the equations of motion:

$$\partial_\alpha(\sqrt{-\gamma}\gamma^{\alpha\beta}\partial_\beta X^\mu) = 0. \tag{1.12}$$

Here $\alpha, \beta = 0, 1$, $\partial_0 = \partial_\tau$ and $\partial_1 = \partial_\sigma$.

1.2.2 Polyakov string

Similarly to what was done in (1.6), we want to introduce a field $h^{\alpha\beta}$, which defines a metric on the world-sheet. The most important example realising this condition is the *Polyakov string*:

$$S = -\frac{T}{2} \int d^2\sigma \sqrt{-h} h^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu}. \tag{1.13}$$

The EOM of the theory for X^μ and $h^{\alpha\beta}$ are:

$$\begin{aligned} &-\frac{1}{2}\sqrt{-h} h_{\alpha\beta} h^{\gamma\rho} \partial_\rho X^\mu \partial_\gamma X^\nu \eta_{\mu\nu} + \sqrt{-h}\partial_\alpha X^\mu \partial_\beta X^\nu = 0, \\ &\partial_\alpha(\sqrt{-h} h^{\alpha\beta} \partial_\beta X^\mu) = 0. \end{aligned} \tag{1.14}$$

In this case one sees that $h^{\alpha\beta} = \gamma^{\alpha\beta}$ is a solution in addition to equation (1.14), but contrary to what was seen in (1.6) it is no longer the only one: the field $h^{\alpha\beta}$ can take other values besides the induced metric, because of further symmetries in the theory.
We observe that the Polyakov action can be interpreted as a classical field theory in 1+1 dimensions (one temporal and one spatial) able to determine the metric $h^{\alpha\beta}$ and D scalar fields X^μ.

1.2.3 Symmetries and gauge invariances

Let us now study the symmetries of (1.13):

- The action is invariant under Poincaré transformations of the type:

$$X^\mu \to \Lambda^\mu_\nu X^\nu + c^\mu. \tag{1.15}$$

  With Λ^μ_ν a Lorentz transformation and c^μ a constant four-vector. We observe that translations of the scalar fields in the background spacetime constitute a continuous symmetry, and by Noether's theorem we will have D conserved charges:

$$\mathbb{P}_\mu = \int_{-R/2}^{+R/2} \mathrm{d}\sigma P_\mu \tag{1.16}$$

  In particular we call energy and angular momentum, respectively, the momenta associated with the spatial coordinates 0 and 1.

$$E = -\int_{-R/2}^{+R/2} \mathrm{d}\sigma P_0 \qquad J = \int_{-R/2}^{+R/2} \mathrm{d}\sigma P_1. \tag{1.17}$$

- The action is invariant under changes of coordinates in (τ, σ), under which the X^μ transform as scalars while $h^{\alpha\beta}$ transforms as a (0,2) tensor:

$$\sigma \to \tilde{\sigma}(\sigma), \qquad \tilde{X}^\mu(\tilde{\sigma}) = X^\mu(\sigma), \quad \tilde{h}_{\alpha\beta}(\tilde{\sigma}) = \frac{\partial\sigma^\gamma}{\partial\tilde{\sigma}^\alpha}\frac{\partial\sigma^\delta}{\partial\tilde{\sigma}^\beta} h_{\gamma\delta}(\sigma). \tag{1.18}$$

- The action is invariant under conformal transformations of the metric:

$$h_{\alpha\beta} \to f(\sigma) h_{\alpha\beta}. \tag{1.19}$$

  Here $f(\sigma)$ is a generic function of the coordinates (up to regularity assumptions). Note that this symmetry is characteristic of the two-dimensional structure of the string, and cannot be generalised in a trivial way to higher dimensions. Indeed, if one imagines generalising the Polyakov action by considering d coordinates for the world-sheet, the transformation (1.19) would lead to:

$$\sqrt{-h} \to f(\sigma)^{d/2}\sqrt{-h}, \qquad h^{\alpha\beta} \to f(\sigma)^{-1} h^{\alpha\beta}. \tag{1.20}$$

  From which it is immediate to say that the function $f(\sigma)$ cancels only if $d = 2$. This invariance of the action is called *Weyl symmetry*, and it indicates that the physics does not depend on possible local dilatations of spacetime that preserve the angles between vectors.

We note that reparametrisation invariance together with Weyl symmetry allow for a lot of freedom in the solutions for the metric $h^{\alpha\beta}$. Indeed one can show that through changes of coordinates it is always possible to make the metric locally flat, i.e. of the form:

$$\tilde{h}_{\alpha\beta} = e^{\phi}\eta_{\alpha\beta}. \tag{1.21}$$

Here $\phi(\sigma)$ is a real function, while $\eta_{\alpha\beta} = \mathrm{diag}(-1, 1)$ is the flat metric. Once this is done, Weyl symmetry allows one to transform $h_{\alpha\beta}$ directly into $\eta_{\alpha\beta}$.
The presence of these local symmetries, i.e. depending on the coordinates of the fields, suggests that the theory contains non-physical or redundant degrees of freedom. In other words, every physical observable is a particular combination of the objects computable from the theory (scalar fields, metrics, tensor fields,...), and it remains invariant under certain local transformations of these objects, called *gauge transformations*. To compute the observables one therefore chooses to impose one or more properties on the computable objects, compatible with such transformations, which eliminate or reduce the redundant degrees of freedom. This procedure is called *gauge fixing*.
The simplest example is obtained by considering the vector potential A^μ and the antisymmetric electromagnetic tensor $F^{\mu\nu}$. Assuming that the entries of $F^{\mu\nu}$ are the observables and that A^μ is defined starting from the relation $\partial_\mu A_\nu - \partial_\nu A_\mu = F_{\mu\nu}$, one easily realises that to each $F^{\mu\nu}$ there corresponds a family of vector potentials, obtaining a gauge invariance of the type:

$$A^\mu_\phi = A^\mu + \partial_\mu \phi. \tag{1.22}$$

Here A^μ is a vector potential and ϕ a generic function. Knowing that the observables do not change as the choice of ϕ varies, we can fix the gauge, at least partially, by imposing a condition on A^μ. In classical electromagnetism the most common gauge choice is the so-called *Lorenz gauge*:

$$\partial_\mu A^\mu = 0. \tag{1.23}$$

Which makes it possible, for example, to write Maxwell's equations in vacuum through the wave equation:

$$\Box A^\mu = 0. \tag{1.24}$$

In general it may not be easy to find a complete and explicit expression for the whole family of solutions of the fields considered, and what one usually has is a set of transformations or symmetries of the fields that allow one to remain within the same family, in the case studied Weyl symmetry and invariance under changes of coordinates. In the next sections we will see that it is possible to detect these additional degrees of freedom in the Polyakov string action, and subsequently how to fix the gauge, making use of the Hamiltonian formalism.

1.2.4 Hamiltonian formalism

Let us rewrite the problem (1.13) in the Hamiltonian formalism. We define the conjugate momenta:

$$P^\mu = \frac{\delta S}{\delta \dot{X}_\mu} = -T\sqrt{-h}h^{00}\dot{X}^\mu - T\sqrt{-h}h^{01}\acute{X}^\mu. \tag{1.25}$$

One can check that the relation between P^μ and $\dot{X}^\mu$ is invertible from:

$$\begin{aligned} &\frac{\partial P^\mu}{\partial \dot{X}^\nu} = -T\sqrt{-h}h^{00}\delta^\mu_\nu, \\ &\det\left(\frac{\partial P^\mu}{\partial \dot{X}^\nu}\right) \neq 0. \end{aligned} \tag{1.26}$$

Inverting:

$$\dot{X}^\mu = -\frac{1}{T\sqrt{-h}h^{00}}P^\mu - \frac{h^{01}}{h^{00}}\acute{X}^\mu. \tag{1.27}$$

The action can be rewritten as:

$$\begin{aligned} S = &-\frac{T}{2}\int \mathrm{d}^2\sigma\sqrt{-h}h^{00}\left(\frac{1}{T\sqrt{-h}h^{00}}P^\mu + \frac{h^{01}}{h^{00}}\acute{X}^\mu\right)^2 \\ &- 2\sqrt{-h}h^{01}\acute{X}_\mu\left(\frac{1}{T\sqrt{-h}h^{00}}P^\mu + \frac{h^{01}}{h^{00}}\acute{X}^\mu\right) + \sqrt{-h}h^{11}(\acute{X}^\mu)^2 \\ = &-\frac{T}{2}\int \mathrm{d}^2\sigma\left(\frac{1}{T^2\sqrt{-h}h^{00}}P^\mu P_\mu - \frac{1}{\sqrt{-h}h^{00}}\acute{X}^\mu\acute{X}_\mu\right). \end{aligned} \tag{1.28}$$

To write the Hamiltonian density:

$$P_\mu\dot{X}^\mu = -\frac{1}{T\sqrt{-h}h^{00}}P^\mu P_\mu - \frac{h^{01}}{h^{00}}P^\mu\acute{X}_\mu, \tag{1.29}$$

$$\begin{aligned} \mathcal{H} = P^\mu\dot{X}_\mu - S &= -\frac{1}{T\sqrt{-h}h^{00}}P_\mu P^\mu - \frac{h^{01}}{h^{00}}P^\mu\acute{X}_\mu + \frac{1}{2T\sqrt{-h}h^{00}}P^\mu P_\mu - \frac{T}{2\sqrt{-h}h^{00}}\acute{X}^\mu\acute{X}_\mu \\ &= -\frac{1}{2T\sqrt{-h}h^{00}}P_\mu P^\mu - \frac{h^{01}}{h^{00}}P^\mu\acute{X}_\mu - \frac{T}{2\sqrt{-h}h^{00}}\acute{X}^\mu\acute{X}_\mu. \end{aligned} \tag{1.30}$$

For the subsequent discussion it is useful to rewrite the action in a form containing both $\dot{X}^\mu$ and P^μ.

$$S = \int \mathrm{d}^2\sigma\left(P^\mu\dot{X}_\mu - \mathcal{H}\right) = \int \mathrm{d}^2\sigma\left(P^\mu\dot{X}_\mu + \frac{1}{2T\sqrt{-h}h^{00}}P^\mu P_\mu + \frac{h^{01}}{h^{00}}P^\mu\acute{X}_\mu + \frac{T}{2\sqrt{-h}h^{00}}\acute{X}^\mu\acute{X}_\mu\right). \tag{1.31}$$

At this point we impose the first gauge-fixing condition. Given what was written in the previous paragraph, thanks to the freedom of changing coordinates and to Weyl symmetry we can impose[5]:

$$\det(h_{\alpha\beta}) = -1. \tag{1.32}$$

With this simplification (1.31) becomes:

$$S = \int \mathrm{d}^2\sigma\left(P^\mu\dot{X}_\mu + \frac{1}{2Th^{00}}(P^2 + T^2\acute{X}^2) + \frac{h^{01}}{h^{00}}P^\mu\acute{X}_\mu\right). \tag{1.33}$$

In the next chapter we will see how the structure of (1.31) was partly predictable a priori, by studying the nature of the constraints arising from the equation of motion of $h^{\alpha\beta}$. From this analysis the condition for fixing the gauge will follow, which will lead to the reduction of the complete system of D fields of the background coordinates + 3 components of the metric to a reduced one of $D-2$ coordinate fields.

[5]This condition is satisfied, for example, when the metric is fixed to be flat in Minkowski space. In what follows, to keep the discussion general (and since this relation cannot be substituted directly into the action), some quantities will be computed without this condition.

Constraints and gauge fixing

2.1 Constrained Hamiltonian systems

Before returning to the discussion of the Polyakov string, in the next two paragraphs we will introduce the theory of the so-called constrained Hamiltonian systems, that is, systems in which constraints of the form $\phi_m(q,p)=0$, to which the solutions must belong, are imposed on the Hamilton-Jacobi equations. The system of constraint equations, under suitable differentiability assumptions, defines a submanifold of phase space, called the *constraint manifold*, to which one wants to restrict the Hamiltonian analysis.

2.1.1 Classification of constraints and gauge freedom

A first classification of Hamiltonian constraints can be made according to how they appear within the theory[6]. In the case in which the condition of non-invertibility of the momenta is satisfied:

$$\det\left(W_{nm}\right) := \det\left(\frac{\partial p_n}{\partial \dot{q}^m}\right) = \det\left(\frac{\partial^2 L}{\partial \dot{q}^n \partial \dot{q}^m}\right) = 0. \tag{2.34}$$

The momenta are not mutually independent and, by the inverse function theorem, the $\dot{q}^n$ cannot be written as functions of the p_n. Given this dependence, we expect to be able to find a relation among the momenta $\phi(q,p)=0$ that makes the non-invertibility explicit.

Def 1. *A non-trivial constraint $\phi(q,p)=0$ is called primary if the following relation holds:*

$$\frac{\partial p_n}{\partial \dot{q}^m}\frac{\partial \phi(q,p)}{\partial p_n} = 0. \tag{2.35}$$

That is, if its gradient with respect to the momenta is an element of the kernel of W_{nm}.

The definition above follows naturally from the chain rule:

$$\phi(q,p)=0 \implies \quad 0 = \frac{\partial \phi(q,p(q,\dot{q}))}{\partial \dot{q}^m} = \frac{\partial \phi(q,p)}{\partial p_n}\frac{\partial p_n}{\partial \dot{q}^m}. \tag{2.36}$$

An example of a primary constraint is given by the dispersion law (1.5), for which one checks that:

$$\phi(X,P) = P^2 + m^2, \qquad \frac{\partial \phi}{\partial P_\mu} = 2P^\mu, \qquad \frac{\partial P_\mu}{\partial \dot{X}_\nu} = m\frac{\delta^\nu_\mu \dot{X}^2 - \dot{X}_\mu \dot{X}^\nu}{(-\dot{X}^2)^{3/2}}, \qquad \frac{\partial P_\mu}{\partial \dot{X}_\nu}\frac{\partial \phi}{\partial P_\mu} = 0. \tag{2.37}$$

From the Euler-Lagrange equations

$$W_{mn}\ddot{q}^n + \frac{\partial^2 \mathcal{L}}{\partial \dot{X}^m \partial X^n}\dot{q}^n - \frac{\partial L}{\partial q^m} = 0, \tag{2.38}$$

it is clear that the vanishing of the determinant of the Jacobian matrix indicates that the second derivative of the positions $\ddot{q}$ is no longer uniquely determined by q and $\dot{q}$. This suggests that the solutions obtained are not uniquely determined by the initial conditions, but contain arbitrary degrees

[6] For the proofs of the theorems not proved here, see [6].

of freedom, to which we want to associate the gauge invariances of the theory. From the Hamiltonian point of view, as already said, condition (2.34) indicates that the $\dot{q}$ will not be invertible with respect to the p. Applying the Legendre transform:

$$H(q,\dot{q}) = \dot{q}^n p_n - L, \tag{2.39}$$

and computing the variation of the Hamiltonian in positions and velocities:

$$\delta H = \dot{q}^n \delta p_n + \delta\dot{q}^n p_n - \delta\dot{q}^n \frac{\partial L}{\partial \dot{q}^n} - \delta q^n \frac{\partial L}{\partial q^n} = \dot{q}^n \delta p_n - \frac{\partial L}{\partial q^n}\delta q^n, \tag{2.40}$$

we observe that the variation of H does not depend on a generic variation in $\delta\dot{q}^n$ but on a combination of them in δp, and for this reason, although $\dot{q}(p)$ cannot be written, it is nevertheless possible to write the Hamiltonian as a function of the momenta $H = H(q,p)$.
Moreover, since the momenta are not all independent, the variation δp is not arbitrary but must respect the condition defined by the constraints:

$$\delta\phi = 0. \tag{2.41}$$

Since the number of equations provided by the calculus of variations is reduced, we expect that there may exist a family of H satisfying (2.40). In particular, what we want to show is that to the Lagrangian system we can associate a family of equivalent Hamiltonians defined by the transformation:

$$H \to H + c^m(q,p)\,\phi_m(q,p). \tag{2.42}$$

The first observation to make is that in the presence of constraints the Hamilton-Jacobi equations are modified. From the computation of the variation (2.40), writing:

$$\delta H = \frac{\partial H}{\partial q^n}\delta q^n + \frac{\partial H}{\partial p_n}\delta p_n, \tag{2.43}$$

one finds that:

$$\left(\frac{\partial H}{\partial q^n} + \frac{\partial L}{\partial q^n}\right)\delta q^n + \left(\frac{\partial H}{\partial p_n} - \dot{q}^n\right)\delta p_n = 0. \tag{2.44}$$

Here δq^n and δp_n are arbitrary variations tangent to the constraint surface. The solution for variations of this type is given by the following lemma:

Lemma 1. *If $\lambda_n\delta q^n + \mu^n\delta p_n = 0$ holds for every variation δq^n and δp_n tangent to the constraint manifold defined by $\phi_m = 0, \quad m = 1,..,M$, then:*

$$\lambda_n = u^m\frac{\partial\phi_m}{\partial q^n},$$
$$\mu^n = u^m\frac{\partial\phi_m}{\partial p_n}.$$

For some functions u^m.

Consequently the solutions to (2.44) turn out to be:

$$\begin{aligned}\dot{q}^n &= \frac{\partial H}{\partial p_n} + u^m\frac{\partial\phi_m}{\partial p_n},\\ \dot{p}_n &= -\frac{\partial H}{\partial q^n} - u^m\frac{\partial\phi_m}{\partial q^n}.\end{aligned} \tag{2.45}$$

This condition allows one to write the time evolution of the functions $F(q,p)$ as:

$$\dot{F} = \{F,H\} + u^m\{F,\phi_m\}. \tag{2.46}$$

The u can be thought of as variables that allow invertibility between the space $(q,\dot{q})$ and the "extended" phase space (q,p,u). The second type of constraint arises from requiring that the constraints remain constant along the equations of motion $\dot{\phi}_n = 0$, described by (2.46).

Def 2. *A Hamiltonian constraint $\varphi(q,p)=0$ is called secondary if it can be written as:*

$$\varphi(q,p)=\{\Phi,H\}+u^m\{\Phi,\phi_m\}. \tag{2.47}$$

Where Φ is a Hamiltonian constraint that must satisfy $\dot{\Phi}=0$, while the ϕ_m are primary Hamiltonian constraints.

An example of a secondary constraint is given by the EOM of $h^{\alpha\beta}$ (1.14). We will say that a set of constraints $\{\Phi_n\}_n$ is complete for a system if the constraints are independent of one another, i.e. such that there is no $\tilde{\Phi}=\sum_{\Phi_n\neq\tilde{\Phi}}c^n\Phi_n$ for some functions $c^n(q,p)$, and if no further independent constraints can be added to this set for the system considered. With this classification, given M primary and N-M secondary constraints, we can write a generic constraint Φ_n as:

$$\Phi_n=\phi_n,\quad n=1,..M;\qquad \Phi_n=\varphi_n,\quad n=M+1,..N. \tag{2.48}$$

Requiring our solutions to lie on the constraint manifold, it makes sense to regard two functions as the "same function" if they differ only by a linear combination of constraints.

Def 3. *Given constraints ϕ_n, two functions $F(q,p)$ and $G(q,p)$ will be said to be weakly equal (written $F\approx G$) if:*

$$F-G=c^n(q,p)\Phi_n. \tag{2.49}$$

With $c^n(q,p)$ certain functions depending on the coordinates.

It is therefore possible to generalise condition (2.47) by requiring only weak equality. Considering a complete set of constraints Φ_n, $n=1,..,N$ we can write:

$$\{\Phi_n,H\}+u^m\{\Phi_n,\phi_m\}\approx 0. \tag{2.50}$$

With ϕ_n, $m=1,..,M$ primary constraints. At fixed q and p the equation above defines a linear system for the unknowns u^m, whose solution will be:

$$u^m=U^m+V^m. \tag{2.51}$$

Here U^m is the particular solution of the system, while V^m is the general solution of the homogeneous equation:

$$V^m\{\Phi_n,\phi_m\}\approx 0,\quad \forall\ n=1,..,N. \tag{2.52}$$

From linear algebra we can write $V^m=v^aV_a^m\quad a=1,..A$, where the v_a are completely arbitrary functions and the V_a^m independent solutions. Equation (2.51) becomes:

$$u^m\approx U^m+v^aV_a^m. \tag{2.53}$$

Substituting u^m in (2.46) we obtain:

$$\begin{aligned}&\dot{F}\approx\{F,H_T\},\\&H_T:=H'+v^a\phi_a,\\&H':=H+U^m\phi_m,\\&\phi_a=V_a^m\phi_m.\end{aligned} \tag{2.54}$$

We therefore note that the equations of motion are determined by a Hamiltonian defined up to arbitrary sums of the primary constraints ϕ_a. This arbitrariness is a symptom of gauge freedom in the full space, which would disappear only if we reduced ourselves to studying the system directly on the constraint manifold determined by the ϕ_a.
We also observe that in this discussion it is not strictly necessary for the constraints to be primary: what matters is that the ϕ_a satisfy $\{\Phi_n,\phi_a\}\approx 0$. This observation leads to a second classification of constraints, which disregards their origin (the discussion and motivation of which is given in the next paragraph).

Def 4. *Given a complete system of constraints* $\{\Phi_n\}_{n=1,..,N}$ *a constraint* Φ_i *is called first-class if:*

$$\{\Phi_i, \Phi_n\} \approx 0 \quad \forall\ n = 1, .., N. \tag{2.55}$$

If there exists an n for which the condition above is not satisfied, one speaks of a second-class constraint.

From the definition one easily deduces that H_T, H' and ϕ_a are all first-class constraints. To make all the gauge invariances explicit it is possible to introduce a Hamiltonian H_E more general than H_T:

$$H_E = H' + u^a \gamma_a. \tag{2.56}$$

Here the γ_a are all the first-class constraints. It is important to stress that H_E is not equivalent to the original Lagrangian system, and its solutions possess additional degrees of freedom. The role of H_E is to make all the gauge invariances of the system explicit. For example, we will show that the Hamiltonian (1.33) is written in a form of the type (2.56), with a particular choice of the coefficients fixed by the metric.
Let us now see how the formulation (2.54) of H_T makes the redundancy of the theory explicit. Since the functions v^a are arbitrary, the time evolution of any function $F(q,p)$ computed using two different sets of functions v^a and v'^a is allowed. Comparing the variation of F with these two rules, for the same initial condition, after an infinitesimal interval δt one obtains that transformations of the type:

$$\delta F = \delta v^a \{F, \phi_a\}. \tag{2.57}$$

With $\delta v^a = (v^a - v'^a)\delta t$ determine a family of acceptable time solutions of F. Transformations like the one above are precisely the infinitesimal gauge transformations we were looking for. They also give an idea of what it means to apply a gauge-fixing condition: since the equation above determines an a-parameter multi-flow of possible solutions, fixing a gauge means choosing B further constraints:

$$C_b(q,p) \approx 0, \quad b = 1, ..B, \tag{2.58}$$

such that they are compatible with the constraints already present, and that they intersect each one-parameter flow generated by the gauge transformations in a single point.
In the next paragraph we will discuss how to eliminate the constraints from the problem completely, reducing to a free system.

2.1.2 Geometric interpretation and gauge fixing

Let us now see how the problem of constrained Hamiltonian systems can be defined through the formalism of symplectic geometry. We begin with some preliminary definitions:

Def 5. *A differentiable manifold* $\mathcal{M}$ *is called symplectic if it is endowed with a closed non-degenerate 2-form* Ω*, called the symplectic form.*

Def 6. *Given a symplectic manifold* $(\mathcal{M}, \Omega)$ *and a scalar function* f*, the Hamiltonian vector field* X_f *is the vector field such that* $\Omega(X_f, \cdot) = df(\cdot)$.

Poisson brackets can be defined as:

$$\{f, g\} = \Omega(X_f, X_g). \tag{2.59}$$

Def 7. *Given two symplectic manifolds* $(\mathcal{M}, \Omega)$ *and* $(\mathcal{N}, \Omega')$*, a function f:* $(\mathcal{M}, \Omega) \to (\mathcal{N}, \Omega')$ *is called a symplectomorphism if* f *is a diffeomorphism and:*

$$f^* \Omega' = \Omega. \tag{2.60}$$

Where f^* *is the pull-back of* f.

When a constraint manifold is considered, in general no symplectic form is induced through the pull-back from the larger manifold; in particular the invertibility of the 2-form may be lost.

Def 8. *A differentiable manifold $\mathcal{M}$ is called presymplectic if it is endowed with a closed non-invertible 2-form, called the presymplectic form.*

The kernel of a presymplectic form is given by the vector fields satisfying the relation:

$$Ker(\Omega) = \{v^i \in TM : \quad \Omega_{ij} v^j = 0\}. \tag{2.61}$$

Starting from a symplectic manifold $(\mathcal{M}, \Omega)$ one can induce on the constraint manifold $\mathcal{C}$ a presymplectic form $\bar{\Omega}$ through the pull-back along the inclusion $\iota : \mathcal{C} \to \mathcal{M}$.

$$\bar{\Omega} = \iota^* \Omega. \tag{2.62}$$

Let us now show that first-class constraints belong to the kernel of $\bar{\Omega}$. Let C^I be a first-class constraint; by definition (4) we have $\{C^I, C^J\} = 0$ along the constraint manifold, for every constraint C^J. Since $dC^J(X_{C^I}) = \{C^J, C^I\} = 0$, the vector field generated by C^I is tangent to the constraint manifold, and it is possible to define its restriction to it $\bar{X}$. It suffices to observe that:

$$\bar{\Omega}_{\alpha\beta} \bar{X}^\beta = \iota^*(\Omega_{ij} X^j_{C^I}) = \iota^*(\partial_i C^I) = 0. \tag{2.63}$$

This proves what was expected. To define a symplectic form on $\mathcal{C}$ it is necessary to take its quotient with respect to the flows generated by the vector fields of the first-class constraints, so as to make the kernel trivial and the presymplectic form $\bar{\Omega}$ non-degenerate. In the formalism seen in the previous section, the Hamiltonian vector field generated by the first-class constraints defines a gauge transformation, while the process of factoring out the flow is equivalent to the gauge-fixing operation.
Once the first-class constraints have been factored out, it is possible to reduce to a reduced Hamiltonian system, in which for all quantities one considers only their restriction to the constraint manifold, through the Dirac brackets:

Def 9. *The Dirac brackets are:*

$$\{f, g\}_D = \{f, g\} - \sum_{ij} \{f, C^i\} (\{C^i, C^j\})^{-1} \{C^j, g\}. \tag{2.64}$$

We observe that:

1. To define the Dirac brackets it is necessary that $\{C^i, C^j\}$ be invertible. The condition is satisfied once the first-class constraints have been factored out;

2. If $g = C^k$ is a second-class constraint we have:

$$\{f, C^k\}_D = \{f, C^k\} - \sum_{ij} \{f, C^i\} (\{C^i, C^j\})^{-1} \{C^j, C^k\} = \{f, C^k\} - \sum_i \{f, C^i\} \delta^k_i = 0. \tag{2.65}$$

   As expected (the restriction of the constraints to the constraint manifold being zero), the vector fields generated by the constraints are tangent to the manifold and give vanishing Dirac brackets with every function f;

3. One can show that the Dirac brackets are induced directly by the pull-back (2.62).

Once the first-class constraints are factored out, the constraint manifold together with the Dirac brackets constitutes the reduced Hamiltonian system we were looking for.
Finally, note that the discussion makes no reference to the distinction between primary and secondary constraints, which justifies what was said about their relevance in the previous paragraph.

2.1.3 Hamiltonian formalism for a field theory in D=2

It is possible to adapt the previous discussion to the case of the string, where a spatial parameter σ is also present, which parametrises its points in spacetime. Since this parameter is space-like, one expects the previous treatment to carry over, up to defining the behaviour of this parameter.
One introduces the Hamiltonian density $\mathcal{H}$, described by the relation:

$$H(\tau) = \int_{-R/2}^{R/2} \mathrm{d}\sigma\, \mathcal{H}(\tau, \sigma). \tag{2.66}$$

The Hamiltonian density can be obtained from the Lagrangian density through the Legendre transform. Working with functions $f = f(\tau, \sigma)$ on phase space, it is possible to define the Poisson brackets by requiring that they be non-zero only when they depend on the same σ, and that at fixed σ they behave like the classical Poisson brackets[7]:

$$\{X^j(\sigma), X_k(\sigma')\} = 0, \qquad \{P^j(\sigma), P_k(\sigma')\} = 0, \qquad \{X^j(\sigma), P_k(\sigma')\} = \delta^j_k \delta(\sigma - \sigma'). \tag{2.67}$$

The Hamilton-Jacobi equations will read:

$$\dot{F}(\tau, \sigma) = \{F, H\} = \int_{-R/2}^{R/2} \mathrm{d}\sigma' \{F(\sigma), \mathcal{H}(\sigma')\}. \tag{2.68}$$

Within this formalism, all the definitions introduced remain valid.

2.2 Constraints for the Polyakov string

Let us apply what we have seen to the Polyakov string, showing that it has two secondary first-class constraints:

$$\begin{cases} C_1(P, \acute{X}) = P^2 + T^2 \acute{X}^2 = 0, \\ C_2(P, \acute{X}) = P^\mu \acute{X}_\mu = 0. \end{cases} \tag{2.69}$$

From the EOM for the metric (1.14) with the condition $h = -1$ one obtains:

$$\frac{h_{\alpha\beta}}{T} \mathcal{L} + \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu} = 0. \tag{2.70}$$

Where we recall that:

$$h_{00} = -h^{11}, \qquad h_{11} = -h^{00}, \qquad h_{01} = h^{01}.$$

Using the Hamiltonian form of $\mathcal{L}$ (1.30) one obtains for $\alpha = \beta = 1$:

$$-\frac{1}{2Th^{00}} \frac{h_{11}}{T} (P^2 - T^2 \acute{X}^2) + \acute{X}^2 = 0 \implies P^2 + T^2 \acute{X}^2 = 0. \tag{2.71}$$

While for $\alpha = 1$, $\beta = 0$:

$$\begin{aligned} &\frac{h_{01}}{T} \left(\frac{1}{2Th^{00}}\right)(P^2 - T^2 \acute{X}^2) + \acute{X}_\mu \left(\frac{1}{h^{00}}\right)\left(\frac{P^\mu}{T} + h^{01} \acute{X}^\mu\right) = 0, \\ &\frac{h^{01}}{2T^2}(P^2 + T^2 \acute{X}^2) + \frac{1}{T} P_\mu \acute{X}^\mu = 0 \implies P_\mu \acute{X}^\mu = 0. \end{aligned} \tag{2.72}$$

The case $\alpha = \beta = 0$ does not provide further information, since the value of one component of the metric has already been determined by fixing the determinant. To show that the constraints are first-class one must compute their mutual Poisson brackets. One obtains:

$$\begin{aligned} \{C_1(\sigma), C_1(\sigma')\} &= 4T^2 \partial_\sigma C_2(\sigma) \delta(\sigma - \sigma') + 8T^2 C_2(\sigma) \partial_\sigma \delta(\sigma - \sigma') \approx 0, \\ \{C_2(\sigma), C_2(\sigma')\} &= \partial_\sigma C_2(\sigma) \delta(\sigma - \sigma') + 2C_2(\sigma) \partial_\sigma \delta(\sigma - \sigma') \approx 0, \\ \{C_1(\sigma), C_2(\sigma')\} &= 2C_1(\sigma) \partial_\sigma \delta(\sigma - \sigma') + \partial_\sigma C_1(\sigma) \delta(\sigma - \sigma') \approx 0. \end{aligned} \tag{2.73}$$

We observe that the Hamiltonian (1.33) appears as a particular case of the form of the extended Hamiltonian (2.56), given the presence of the two secondary first-class constraints.

[7] For a more formal and complete treatment, the operations should be understood in a functional sense.

2.3 Uniform light-cone gauge

Having found two first-class constraints, we expect to have to impose two gauge-fixing conditions on the system. To fix the gauge it is convenient to perform a change of variables of the scalar fields X^μ depending on a parameter a, passing to *light-cone coordinates*.

$$\begin{cases} X^+ = aX^0 + (1-a)X^1, \\ X^- = -X^0 + X^1. \end{cases} \tag{2.74}$$

Imposing the Poisson bracket relations (2.67), the transformations for the momenta can be determined:

$$\begin{cases} P_+ = P_0 + P_1, \\ P_- = -(1-a)P_0 + aP_1. \end{cases} \tag{2.75}$$

This transformation induces a transformation of the background spacetime metric $\tilde{\eta}_{\alpha\beta}$, of the form:

$$\begin{aligned} &\tilde{\eta}_{01} = \tilde{\eta}_{10} = 1, \qquad \tilde{\eta}_{00} = 0, \qquad \tilde{\eta}_{11} = 2a-1, \qquad \tilde{\eta}_{jk} = \delta_{jk}, \quad \forall j,k = 2,..,D-1. \\ &\tilde{\eta}^{01} = \tilde{\eta}^{10} = 1, \qquad \tilde{\eta}^{00} = 1-2a, \qquad \tilde{\eta}^{11} = 0, \qquad \tilde{\eta}^{jk} = \delta^{jk}, \quad \forall j,k = 2,..,D-1. \end{aligned}$$

At this point it is possible to fix the so-called *uniform light-cone gauge*:

$$\begin{aligned} X^+ &= \tau, \\ P_- &= 1. \end{aligned} \tag{2.76}$$

The field X^+ is made to coincide with the time parameter τ, and the momentum conjugate to the field X^- is made constant, without explicitly imposing constraints on the variables X^j. For example, it is immediate that the following identities hold:

$$\dot{X}^+ = 1, \qquad \acute{X}^+ = 0. \tag{2.77}$$

If we compute the charge corresponding to P_- we obtain:

$$\int_{-R/2}^{+R/2} \mathrm{d}\sigma P_- = R = E + a(J-E). \tag{2.78}$$

That is, the volume of the world-sheet theory, in this case R, by depending on the parameter a depends on the choice of gauge. We now want to reduce to a free reduced Hamiltonian system of $D-2$ fields X^j, using the constraints to determine the fields $X^\pm$ and $h^{\alpha\beta}$. Starting from the first-class constraints (2.69):

$$P^\mu \acute{X}_\mu = 0 \implies \acute{X}^- = -P_j \acute{X}^j. \tag{2.79}$$

$$\begin{aligned} &P^\mu P_\mu + T^2 \acute{X}^\mu \acute{X}_\mu = 0 \implies (1-2a)P_+^2 + 2P_+ + P_j^2 + T^2 \acute{X}_j^2 + (2a-1)T^2 P_j P_k \acute{X}^j \acute{X}^k = 0, \\ &P_{+,12} = \frac{-1 \pm \sqrt{1-(1-2a)(P_j^2 + T^2 X_j'^2) + (2a-1)^2 T^2 P_j P_k X'^j X'^k}}{1-2a}. \end{aligned} \tag{2.80}$$

Where $P_j^2 = P_i P_j \delta^{ij}$, $\acute{X}_j^2 = \acute{X}_i \acute{X}_j \delta^{ij}$. For the Hamiltonian density to be positive definite, the solution with the '+' sign is chosen:

$$P_+ = \frac{-1 + \sqrt{1-(1-2a)(P_j^2 + T^2 X_j'^2) + (2a-1)^2 T^2 P_j P_k X'^j X'^k}}{1-2a}. \tag{2.81}$$

The first-class constraints impose conditions on the canonical variables conjugate to those fixed by the gauge.

From the first constraint (2.79), imposing periodicity of the spatial coordinate (since we are dealing with a closed string):

$$0 = X^-(R/2) - X^-(-R/2) = \int_{-R/2}^{+R/2} \mathrm{d}\sigma\, \acute{X}^- = -\int_{-R/2}^{+R/2} \mathrm{d}\sigma\, P_i \acute{X}^i. \tag{2.82}$$

We observe that the last term is the Noether charge conserved by the invariance under spatial translations along the string [8]:

$$\sigma \to \tilde{\sigma} = \sigma + c, \quad \tilde{X}^i(\tilde{\sigma}) = X^i(\sigma), \quad Q = \int_{-R/2}^{+R/2} \mathrm{d}\sigma \left(\delta\tau\mathcal{L} - \frac{\partial\mathcal{L}}{\partial(\partial_\alpha \dot{X}^i)}\acute{X}^i\delta\sigma\right) = -c\int_{-R/2}^{+R/2} \mathrm{d}\sigma P_i\acute{X}^i. \tag{2.83}$$

By analogy, just as momentum is the charge conserved by translation invariance, we can say that the conserved charge found is the total momentum of the world-sheet p_{ws}. From the periodicity condition (2.82) one obtains the *level matching condition*:

$$p_{ws} = 0. \tag{2.84}$$

Let us now write the Hamiltonian density of the reduced system. Starting from (1.33), the last two terms, proportional to the constraints of the system, can be dropped:

$$\begin{aligned} S = &\int d^2\sigma P_\mu \dot{X}^\mu = \\ &\int d^2\sigma P_- \dot{X}^- + P_+\dot{X}^+ + P_j\dot{X}^j = \\ &\int d^2\sigma P_j\dot{X}^j + P_+. \end{aligned} \tag{2.85}$$

Hence:

$$\mathcal{H}(X^j, \acute{X}^j, P_j) = P_j\dot{X}^j - \mathcal{L} = -P_+. \tag{2.86}$$

Where the term $\dot{X}^-$ cancels because of the periodicity condition along the string. In the particular case $a = 1/2$ the Hamiltonian density is simply given by:

$$\mathcal{H}(X_j, \acute{X}_j, P_j) = \frac{1}{2}(P_j^2 + T^2\acute{X}_j^2). \tag{2.87}$$

While if $a \neq 1/2$ we have:

$$\mathcal{H} = \frac{1 - \sqrt{1 - (1-2a)(P_j^2 + \acute{X}_j^2) + (1-2a)^2 P_j P_k \acute{X}^j\acute{X}^k}}{1-2a}. \tag{2.88}$$

We have thus obtained the reduced Hamiltonian system we were looking for.
Integrating the Hamiltonian density one obtains the Hamiltonian and its relation with the volume of the theory:

$$H = -\int_{-R/2}^{+R/2} \mathrm{d}\sigma\, P_+ = E - J, \qquad R = J + (1-a)H. \tag{2.89}$$

Finally, let us see how it is possible to determine the metric $h^{\alpha\beta}$. From the definition of the momenta $P_\pm$:

$$\begin{aligned} P_- =& -T\frac{\delta}{\delta\dot{X}^-}(h^{01}\dot{X}^-\acute{X}^-\eta_{--} + 2h^{01}\dot{X}^-\acute{X}^+\eta_{-+} + h^{00}\dot{X}^+\dot{X}^-\eta_{+-} + \frac{1}{2}h^{00}\dot{X}^-\dot{X}^-\eta_{--}) = \\ =& -T(h^{00} + (2a-1)(h^{00}\dot{X}^- + h^{01}\acute{X}^-)) = 1. \end{aligned} \tag{2.90}$$

[8] Note that invariance under time translations yields the conservation of the Hamiltonian.

$$\begin{aligned} P_+ =& -T\frac{\delta}{\delta\dot{X}^+}(h^{01}\dot{X}^+\acute{X}^+\eta_{++} + 2h^{01}\dot{X}^+\acute{X}^-\eta_{+-} + h^{00}\dot{X}^+\dot{X}^-\eta_{+-} + \frac{1}{2}h^{00}\dot{X}^+\dot{X}^+\eta_{++}) = \\ =& -T(2h^{01}\acute{X}^- + h^{00}\dot{X}^-). \end{aligned} \tag{2.91}$$

Adding the equation $h = -1$ and $\partial_\alpha(h^{\alpha\beta}\partial_\beta X^\pm) = 0$ we must solve the following system:

$$\begin{cases} h^{11}h^{00} - h^{10}h^{01} = -1, \\ 1 = -Th^{00} + (2a-1)(P_+ + Th^{01}\acute{X}^-), \\ \partial_0 h^{00} + \partial_1 h^{10} = 0, \\ -\partial_0(\frac{P_+}{T} - h^{01}\acute{X}^-) + \partial_1(h^{11}\acute{X}^- + h^{10}\dot{X}^-) = 0. \end{cases} \tag{2.92}$$

Since the general analytic solution of this system is difficult, we will compute the perturbative expansion of the metric up to second order in the fields X^j. The first observation to make is that for the case $a = 1/2$ the system admits an explicit solution:

$$h^{\alpha\beta} = \begin{pmatrix} -T^{-1} & 0 \\ 0 & T \end{pmatrix}. \tag{2.93}$$

For the case $a \neq 1/2$ we observe that:

- In the case $a = 1/2$ we go back to the result (2.93), in which the metric is constant;
- X'^-, $\dot{X}^-$ and P_+ depend on even powers of the fields, and the order-0 term vanishes;
- the term $2a - 1$ is coupled to a term at least quadratic in the fields.

These observations suggest trying an ansatz of the type:

$$\begin{cases} h^{00} = -T^{-1} + (2a-1)H^{00} + O(X^4), \\ h^{11} = T + (2a-1)H^{11} + O(X^4), \\ h^{10} = (2a-1)H^{10} + O(X^4). \end{cases} \tag{2.94}$$

Where $H^{\alpha\beta}$ is the term of the metric $h^{\alpha\beta}$ quadratic in the fields. From the first condition, on the determinant, one finds:

$$\begin{aligned} h^{00}h^{11} - h^{01}h^{01} &= -1, \\ T^2H^{00} &= H^{11}. \end{aligned} \tag{2.95}$$

From the second equation:

$$\begin{aligned} 1 &= -Th^{00} + (2a-1)(P_+ + Th^{01}\acute{X}^-), \\ H^{00} &= -\frac{(2a-1)}{2T}(P_j^2 + T^2\acute{X}_j^2). \end{aligned} \tag{2.96}$$

Finally, with the third one obtains:

$$\begin{aligned} \partial_0 h^{00} + \partial_1 h^{10} &= 0, \\ H^{10} &= (2a-1)TP^j\acute{X}_j. \end{aligned} \tag{2.97}$$

One can check that, with the solutions above, the last equation of (2.92) becomes an identity. Putting together what we have found, we obtain the metric written up to second order in the fields X^j:

$$h^{\alpha\beta} = \begin{pmatrix} -T^{-1} + \frac{(1-2a)}{2T}(P_j^2 + T^2\acute{X}_j^2) & (2a-1)TP^j\acute{X}_j \\ (2a-1)TP^j\acute{X}_j & T + \frac{T(1-2a)}{2}(P_j^2 + T^2\acute{X}_j^2) \end{pmatrix} + O(X^4). \tag{2.98}$$

Another way to eliminate the degrees of freedom of the metric consists in writing:

$$S = -\frac{T}{2}\int \mathrm{d}^2\sigma\sqrt{-h}h^{\alpha\beta}A_{\alpha\beta}. \tag{2.99}$$

Where $A_{\alpha\beta} = \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu}$, and solving the EOM of the metric:

$$h_{\alpha\beta} = C(\sigma) A_{\alpha\beta}, \qquad C(\sigma) = \frac{2}{h^{\rho\delta} A_{\rho\delta}}. \tag{2.100}$$

From the condition $h = -1$, if $A = \det(A_{\alpha\beta})$ we obtain:

$$\begin{aligned} &\sqrt{-A} = 1/C(\sigma), \\ &h^{\alpha\beta} = \frac{A^{\alpha\beta}}{\sqrt{-A}}, \\ &S = -T \int \mathrm{d}^2\sigma \, \sqrt{-A}. \end{aligned} \tag{2.101}$$

We thus find that, by simplifying the metric, the Nambu-Goto string action is recovered.

The scattering S-matrix

3.1 Expansion of the Hamiltonian

In this chapter, following the approach adopted in [3], we will study the dynamics of the system once the gauge invariance has been removed, through a perturbative expansion. To do so it is convenient to redefine the fields by coupling them to a parameter which, for the expansion, can be assumed to be small. We begin by removing the parameter T through the transformation:

$$\sigma \to \sqrt{T}\sigma, \qquad X^i \to \frac{X^j}{\sqrt{T}}, \qquad P^j \to \frac{P^j}{\sqrt{T}}. \tag{3.102}$$

After the rescaling, looking at the Hamiltonian density (2.88) it is clear that the coupling parameter we are looking for is $\beta := \frac{1-2a}{T}$. At this point we can write the perturbative expansion of $\mathcal{H}$ up to the term quartic in the fields, or equivalently linear in β:

$$\mathcal{H} = \frac{1}{2}\left(P_j^2 + \acute{X}_j^2\right) + \frac{\beta}{8}\left(P_j^2 P_k^2 + 2P_j^2 \acute{X}_k^2 + \acute{X}_j^2 \acute{X}_k^2 - 4P_j P_k \acute{X}^j \acute{X}^k\right) + O(\beta^2). \tag{3.103}$$

3.1.1 Quadratic term: EOM

Assuming $\beta \ll 1$ we can write $\mathcal{H} = \mathcal{H}^{(2)} + \beta\mathcal{H}^{(4)} + o(\beta^2)$, and compute the equations of motion starting from $\mathcal{H}_0 = \mathcal{H}^{(2)}$.

$$\mathcal{H}_0 = \frac{1}{2}(P_j^2 + \acute{X}_j^2). \tag{3.104}$$

From the Hamilton-Jacobi equations one obtains:

$$\begin{cases} \dot{X}_j = P_j, \\ \dot{P}_j = X_j''. \end{cases} \tag{3.105}$$

$$\ddot{X}_j - X_j'' = 0. \tag{3.106}$$

Where (3.106) is a wave equation and has solution:

$$\begin{aligned} X_j(\sigma,\tau) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\mathrm{d}p}{\sqrt{2\omega}} \left(a_j(p) e^{ip\sigma - i\omega t} + a_j^*(p) e^{-ip\sigma + i\omega t}\right), \\ \acute{X}_j(\sigma,\tau) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\mathrm{d}p}{\sqrt{2\omega}} ip \left(a_j(p) e^{ip\sigma - i\omega t} - a_j^*(p) e^{-ip\sigma + i\omega t}\right), \\ P_j(\sigma,\tau) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{\mathrm{d}p}{\sqrt{2\omega}} (-i\omega) \left(a_j(p) e^{ip\sigma - i\omega t} - a_j^*(p) e^{-ip\sigma + i\omega t}\right). \end{aligned} \tag{3.107}$$

Where the $a_j(p)$ are arbitrary functions of p, while ω is given by the dispersion relation:

$$\omega = \sqrt{p^2}. \tag{3.108}$$

We observe that the dispersion relation (3.108) is not analytic at $p = 0$, because of the presence of the square root and the absence of a mass term (present, for example, for relativistic particles). This non-analyticity generates problems at low energies, called *infrared problems*.
Having solved the equations for the fields, we rewrite the unperturbed Hamiltonian in terms of a and a^*:

$$H_0 = \int_{-R/2}^{+R/2} \mathrm{d}\sigma\, \mathcal{H}_0. \tag{3.109}$$

Where R is the radius of the world-sheet. To proceed with the computation it is necessary to perform the so-called *decompactification limit*, in which one lets $J \to \infty$, hence also $R \to \infty$. The geometric interpretation of this procedure is that, by sending the radius of curvature to infinity, the world-sheet becomes topologically equivalent to a plane, thus removing all the potential problems related to compactness, such as the impossibility of defining free particles since they are always "finitely far apart".
Applying this approximation one obtains:

$$H_0 = \sum_{j=2}^{D-1} \int_{-\infty}^{\infty} \mathrm{d}p\ \omega(p) a_j^*(p) a_j(p). \tag{3.110}$$

We note that what we have just found is equivalent, up to constants, to the Hamiltonian of $D-2$ decoupled quantum harmonic oscillators defined by the raising operators a_j^* and frequency ω.

3.1.2 Quartic term: scattering S-matrix

The first perturbative term coupled to β can be used to compute the tree-level 2+2 momenta Scattering matrix **S**.
In Scattering theory the **S** matrix is a unitary operator that maps the free-particle *out-states* into the *in-states* in the Heisenberg picture. The in and out states belong to the same Hilbert space of the model, and are eigenvectors of the Hamiltonian H with the same eigenvalue E:

$$H|p_1, p_2, .., p_n\rangle^{(in/out)}_{i_1,i_2,..,i_n} = E|p_1, p_2, .., p_n\rangle^{(in/out)}_{i_1,i_2,..,i_n}. \tag{3.111}$$

Where $i_1, .., i_n$ are indices used to account for the different properties of the particles with momenta $p_1, .., p_n$.
The eigenvalue E can be rewritten as:

$$E = \sum_{k=1}^{n} \omega^{(i_k)}_{p_k}. \tag{3.112}$$

Where $\omega^{(i)}_p$ is the energy given by the dispersion relation of the particle of type i and momentum p.
To describe the in and out states we define in and out creation and annihilation operators satisfying the canonical commutation relations. Let $|\Omega\rangle$ be the vacuum state, which is annihilated:

$$a_{in}(p,t)|\Omega\rangle = a_{out}(p,t)|\Omega\rangle = 0. \tag{3.113}$$

The free in and out states correspond to the states obtained by applying the in creation operators $a_k^{in\,\dagger}(p) := a_k^{in\,\dagger}(p,0)$ and the out ones $a_k^{out\,\dagger}(p) := a_k^{out\,\dagger}(p,0)$ to the vacuum.

$$\begin{aligned} |p_1, p_2, .., p_n\rangle^{(in)}_{i_1,i_2,..,i_n} &= a_{i_1}^{in\,\dagger}(p_1)...a_{i_n}^{in\,\dagger}(p_n)|\Omega\rangle, \\ |p_1, p_2, .., p_n\rangle^{(out)}_{i_1,i_2,..,i_n} &= a_{i_1}^{out\,\dagger}(p_1)...a_{i_n}^{out\,\dagger}(p_n)|\Omega\rangle. \end{aligned} \tag{3.114}$$

Since the in and out operators satisfy the canonical commutation relations, by the Stone - von Neumann theorem they are related by a unitary operator **S**, which we define to be the *Scattering matrix*.

$$a_{in}^\dagger(p,t) = \mathbf{S} a_{in}^\dagger(p,t)\mathbf{S}^\dagger, \quad a_{in}(p,t) = \mathbf{S} a_{in}(p,t)\mathbf{S}^\dagger, \quad \mathbf{S}|\Omega\rangle = |\Omega\rangle. \tag{3.115}$$

The **S** matrix does not depend on time, since the dependence between in and out states cancels out. Consequently, the in and out states are related as follows:

$$|p_1, p_2, .., p_n\rangle^{(in)}_{i_1,i_2,..,i_n} = \mathbf{S}|p_1, p_2, .., p_n\rangle^{(out)}_{i_1,i_2,..,i_n}. \tag{3.116}$$

In particular, for two-particle states we obtain:

$$|p_1, p_2\rangle^{(in)}_{i,j} = \mathbf{S}^{kl}_{ij}|p_1, p_2\rangle^{(out)}_{k,l}, \tag{3.117}$$

considering momenta that do not change during the scattering, and ordering them so that $p_1 > p_2 > ... > p_n$. It is clear that the **S** matrix commutes with the Hamiltonian, and that without interactions it must coincide with the identity. To compute the **S** matrix with perturbation theory, the Hamiltonian is split into a free part plus an interaction part.

$$H = H_0 + V. \tag{3.118}$$

Where the free part depends on the creation and annihilation operators.

$$H_0 = \sum_k \int_{-\infty}^{+\infty} \mathrm{d}p\ \omega_p^{(k)} a_k^\dagger(p,t) a^k(p,t). \tag{3.119}$$

Where the operators a, $a^\dagger$ are in the Heisenberg picture and satisfy the canonical commutation laws:

$$[a_i^\dagger, a^j] = \delta_i^j, \tag{3.120}$$

and obey the equations of motion:

$$\dot{a}^k(p,t) = i[H, a^k(p,t)] = -i\,\omega_p^{(k)} a^k(p,t) + i\,[V, a^k(p,t)]. \tag{3.121}$$

Where $V = V(a^\dagger, a)$. Since a, $a^\dagger$ satisfy the canonical commutation relations, again by the Stone - von Neumann theorem these operators are related to the in and out creation and annihilation operators by unitary transformations.

$$\begin{aligned} a^\dagger(p,t) &= U_{in}^\dagger(t)\, a_{in}^\dagger(p,t)\, U_{in}(t), \qquad a(p,t) = U_{in}^\dagger(t)\, a_{in}(p,t)\, U_{in}(t), \\ a^\dagger(p,t) &= U_{out}^\dagger(t)\, a_{out}^\dagger(p,t)\, U_{out}(t), \qquad a(p,t) = U_{out}^\dagger(t)\, a_{out}(p,t)\, U_{out}(t). \end{aligned} \tag{3.122}$$

Where U_{in} and U_{out} are determined up to constant unitary matrices, which can be fixed by the boundary conditions:

$$U_{in}(-\infty) = \mathbb{I}, \qquad U_{out}(+\infty) = \mathbb{I}. \tag{3.123}$$

And up to a phase $U(t) \to e^{i\varphi(t)}U(t)$. Condition (3.123) makes it possible to interpret the in states as the asymptotic states at $t \to -\infty$ and the out states as the asymptotic states at $t \to +\infty$. Comparing the definitions (3.116) and (3.122) one arrives at:

$$\mathbf{S} = U_{in}(t)\, U_{out}(t). \tag{3.124}$$

Being time-independent, the equation above holds $\forall\ t$, and consequently we can have:

$$\mathbf{S} = U_{in}(+\infty) = U_{out}(-\infty). \tag{3.125}$$

Differentiating and manipulating (3.122) a little, we obtain the equalities:

$$\left[\dot{U}_{in}U_{in}^\dagger + iV(a_{in}^\dagger, a_{in}), a_{in}^\dagger(p,t)\right] = 0, \qquad \left[\dot{U}_{in}U_{in}^\dagger + iV(a_{in}^\dagger, a_{in}), a_{in}(p,t)\right] = 0. \tag{3.126}$$

Which implies $\dot{U}_{in}U_{in}^\dagger + iV(a_{in}^\dagger, a_{in}) = c(t)\mathbb{I}$, where $c(t)$ does not depend on the creation and annihilation operators. By suitably choosing the phase $\varphi(t)$ the term $c(t)$ can be removed, arriving at the equation:

$$\dot{U}_{in}U_{in}^\dagger + iV(a_{in}^\dagger, a_{in}) = 0. \tag{3.127}$$

Which can be solved as:

$$U_{in}(t) = \mathcal{T}\text{exp}\left(-i\int_{-\infty}^{t} \mathrm{d}\tau\, V(a_{in}^{\dagger}(\tau), a_{in}(\tau))\right). \tag{3.128}$$

Where $\mathcal{T}$ denotes the time-ordered operator, the prescription according to which a product of operators is to be understood as ordered in the time variable τ. Similarly for U_{out}:

$$\dot{U}_{out}U_{out}^{\dagger} - iV(a_{out}^{\dagger}, a_{out}) = 0, \tag{3.129}$$

$$U_{in}(t) = \mathcal{T}\text{exp}\left(-i\int_{t}^{\infty} \mathrm{d}\tau\, V(a_{out}^{\dagger}(\tau), a_{out}(\tau))\right). \tag{3.130}$$

Putting together what we have obtained, we can write the Scattering matrix as:

$$\mathbf{S} = \mathcal{T}\text{exp}\left(-i\int_{-\infty}^{+\infty} \mathrm{d}\tau\, V(a^{\dagger}(\tau), a(\tau))\right). \tag{3.131}$$

At this point, thinking of V as coupled to a parameter ϵ, it is possible to perform a perturbative expansion:

$$\mathbf{S} = \mathbb{I} - i\int_{-\infty}^{+\infty} \mathrm{d}\tau\, V(\tau) + O(\epsilon^2). \tag{3.132}$$

In the case of the system under examination we have the perturbative parameter β, coupled linearly to the Hamiltonian containing the fields of quartic order $\mathcal{H}^{(4)}$. We therefore have:

$$\mathbf{S} = \mathbb{I} - i\beta\int \mathrm{d}^2\sigma\, \mathcal{H}^{(4)}(\tau, \sigma) + O(\beta^2) \tag{3.133}$$

Defining the transfer matrix $\mathbf{T}$ through the relation:

$$\mathbf{S} = \mathbb{I} + i\mathbf{T}, \tag{3.134}$$

$$\mathbf{T} = -\beta\int \mathrm{d}^2\sigma\, \mathcal{H}^{(4)}(\tau, \sigma) + O(\beta^2). \tag{3.135}$$

Substituting the quartic term of the Hamiltonian density one obtains:

$$\mathbf{T} = -\frac{\beta}{8}\int \mathrm{d}^2\sigma\, \left(P_j^2P_k^2 + 2P_j^2\acute{X}_k^2 + \acute{X}_j^2\acute{X}_k^2 - 4P_jP_k\acute{X}^j\acute{X}^k\right) + O(\beta^2). \tag{3.136}$$

3.2 S-matrix for the effective action

In this section we want to compute explicitly the transfer matrix given by (3.136). To do so we will compute the **S** matrix for a more general set of systems, which share certain properties with the Polyakov string action. We begin by trying to write an action of $D-2$ scalar fields with the following properties:

- It is Lorentz invariant with respect to the world-sheet metric;
- The fields have $\mathrm{SO}(D-2)$ invariance;
- It is at most quartic in the fields X^j;
- At most first derivatives of the fields X^j appear.

We find that the most general form can be expressed as [7]:

$$\begin{aligned} S &= -\int d^2\sigma\, \mathcal{L}, \\ \mathcal{L} &= \mathcal{L}^{(2)} + \mathcal{L}^{(4)}, \\ \mathcal{L}^{(2)} &= \frac{1}{2}\partial_\alpha X^j \partial^\alpha X^j, \\ \mathcal{L}^{(4)} &= \frac{1}{4}c_1 \partial_\alpha X^k \partial^\alpha X^k \partial_\beta X^j \partial^\beta X^j + \frac{1}{4}c_2 \partial_\alpha X^k \partial^\alpha X^j \partial_\beta X^j \partial^\beta X^k. \end{aligned} \tag{3.137}$$

Where c_1 and c_2 are arbitrary real constants. These effective actions can be interpreted as the possible actions obtained by changing the gauge-fixing choice for an action that is Lorentz invariant also in the X^μ, expanded and truncated at quartic order in the fields. Let us rewrite the derivatives of the fields and the Lagrangian density in terms of the four-momentum:

$$\begin{aligned} \dot{X}_j &= P_j + (c_1+c_2)P_j P_k^2 - c_1 P_j \acute{X}_k^2 - c_2 \acute{X}_j P_k \acute{X}_k, \\ \dot{X}_j P_j &= P_j^2 + (c_1+c_2)P_j^2 P_k^2 - c_1 P_j^2 \acute{X}_k^2 - c_2 P_j \acute{X}_j P_k \acute{X}_k, \\ \mathcal{L}^{(2)} &= \frac{1}{2}\acute{X}_j^2 - \frac{1}{2}P_j^2 - (c_1+c_2)P_j^2 P_k^2 + c_1 P_j^2 \acute{X}_k^2 + c_2 P_j \acute{X}_j P_k \acute{X}_k, \\ \mathcal{L}^{(4)} &= \frac{1}{4}(c_1+c_2)(P_j^2 P_k^2 + \acute{X}_j^2 \acute{X}_k^2) - \frac{1}{2}c_1 P_j^2 \acute{X}_k^2 - \frac{1}{2}c_2 (P_j \acute{X}_j)^2. \end{aligned} \tag{3.138}$$

Finally, the Hamiltonian density is written from the Legendre transform:

$$\mathcal{H} = \frac{1}{2}(P_j^2 + \acute{X}_j^2) + a_1 P_j^2 P_k^2 + a_2 \acute{X}_j^2 \acute{X}_k^2 + a_3 P_j^2 \acute{X}_k^2 + a_4 P_j \acute{X}_j P_k \acute{X}_k. \tag{3.139}$$

With:

$$a_1 = \frac{1}{4}(c_1+c_2), \qquad a_2 = \frac{1}{4}(c_1+c_2), \qquad a_3 = -\frac{1}{2}c_1, \qquad a_4 = -\frac{1}{2}c_2. \tag{3.140}$$

In particular we observe that for the uniform light cone gauge:

$$c_1 = \frac{2a-1}{2}, \qquad c_2 = 1-2a, \qquad c_2 = -2c_1. \tag{3.141}$$

We now move on to the computation of the $\mathbf{T}$ matrix in the general case:

$$\mathbf{T} = -\int d^2\sigma \left(a_1 P_j^2 P_k^2 + a_2 \acute{X}_j^2 \acute{X}_k^2 + a_3 P_j^2 \acute{X}_k^2 + a_4 P_j \acute{X}_j P_k \acute{X}_k \right). \tag{3.142}$$

Integrating in σ and τ, three different products of Dirac deltas appear; of these, only the one of the following form does not vanish:

$$\delta(p_1+p_2-p_3-p_4)\delta(\omega_1+\omega_2-\omega_3-\omega_4) = J(\omega_1,\omega_2)[\delta(p_1-p_3)\delta(p_2-p_4)+\delta(p_1-p_4)\delta(p_2-p_3)]. \tag{3.143}$$

If we assume $p_1 > p_2$ and $p_3 > p_4$ it reduces to:

$$J(\omega_1,\omega_2)\delta(p_1+p_2-p_3-p_4)\delta(\omega_1+\omega_2-\omega_3-\omega_4) = J(\omega_1,\omega_2)\delta(p_1-p_3)\delta(p_2-p_4). \tag{3.144}$$

Where:

$$J(\omega_1,\omega_2) = \frac{\omega_1\omega_2}{|p_1\omega_2 - p_2\omega_1|}. \tag{3.145}$$

Having defined $|p_a,p_b\rangle_{i,j} = a_i^\dagger(p_a)a_j^\dagger(p_b)|0\rangle$ we can write:

$$\mathbf{T}_{ij;kl} = \langle p_1,p_2|\mathbf{T}|p_3,p_4\rangle_{ij;kl} = \frac{1}{4\sqrt{\omega_1\omega_2\omega_3\omega_4}} J(\omega_1,\omega_2) M_{ij;kl}(p_1,p_2,p_3,p_4)\delta(p_1-p_3)\delta(p_2-p_4). \tag{3.146}$$

Where M is called the scattering amplitude. From the $SO(D-2)$ symmetry of the Hamiltonian one observes that the scattering amplitude can be decomposed into only three contributions:

$$M_{k,l}^{i,j}(p_1,p_2,p_3,p_4) = A(p_1,p_2,p_3,p_4)\delta_{k,l}^{i,j} + B(p_1,p_2,p_3,p_4)\delta_{j,l}^{i,k} + C(p_1,p_2,p_3,p_4)\delta_{j,k}^{i,l}. \tag{3.147}$$

Where A can be interpreted as an annihilation term, B as a transition term and C as a reflection term. Computing the term A as an example, because of the structure of the deltas all and only the terms of the form $2a_j(p_{i_1})a_j(p_{i_2})a_k^*(p_{i_3})a_k^*(p_{i_4})$ will contribute, with i_1, i_2, i_3 and i_4 taking the values 1,...,4 and j and k repeated indices from 1 to D-2. To apply the Bras and Kets of the asymptotic states it is necessary to quantise the system, thinking of the a_j as operators and transforming $a^* \to a^\dagger$. Finally, to remove the ordering ambiguity one applies *normal ordering*, in which one conventionally chooses to keep the $a^\dagger$ on the left and the a on the right when they appear multiplied together. With this rule the 4 terms contribute as:

$$\begin{aligned} A = & -8a_1\omega_1\omega_2\omega_3\omega_4 - 8a_2p_1p_2p_3p_4 - 4a_3(p_1p_2\omega_3\omega_4 + p_3p_4\omega_1\omega_2) - \\ & - 2a_4(p_1p_4\omega_2\omega_3 + p_2p_3\omega_1\omega_4 + p_1p_3\omega_2\omega_4 + p_2p_4\omega_1\omega_3) = \\ = & -2c_1(p_1\cdot p_2)(p_3\cdot p_4) - c_2(p_1\cdot p_3)(p_2\cdot p_4) - c_2(p_1\cdot p_4)(p_2\cdot p_3). \end{aligned} \tag{3.148}$$

where $p_a \cdot p_b = -\omega_a\omega_b + p_ap_b$ denotes the Lorentz-invariant product. Now defining the Mandelstam variables:

$$\begin{aligned} s &= -(p_1+p_2)^2 = -2p_1\cdot p_2, \\ t &= -(p_1-p_3)^2 = +2p_1\cdot p_3, \\ u &= -(p_1-p_4)^2 = +2p_1\cdot p_4. \end{aligned} \tag{3.149}$$

Observing that:

$$s+t+u=0, \tag{3.150}$$

and computing B and C in a similar way, it is possible to write them as:

$$A = -\frac{1}{4}(2c_1+c_2)s^2 + \frac{1}{2}c_2tu, \qquad B = -\frac{1}{4}(2c_1+c_2)t^2 + \frac{1}{2}c_2su, \qquad C = -\frac{1}{4}(2c_1+c_2)u^2 + \frac{1}{2}c_2st. \tag{3.151}$$

As a consequence of the ordering convention adopted and of the application of the delta, we have that:

$$p_1 = p_3, \qquad p_2 = p_4, \qquad t = 0, \qquad u = -s. \tag{3.152}$$

Simplifying A, B and C:

$$A = -\frac{1}{4}(2c_1+c_2)s^2, \qquad B = -\frac{1}{2}c_2s^2, \qquad C = -\frac{1}{4}(2c_1+c_2)s^2. \tag{3.153}$$

In the case of the string system with the choice of uniform light-cone gauge, thanks to the relation between the coefficients $2c_1 + c_2 = 0$, the relation above simplifies further:

$$A = 0, \qquad B = -\frac{1}{2}c_2s^2, \qquad C = 0. \tag{3.154}$$

For $J(\omega_1, \omega_2)$ to be well defined it is necessary to impose a second condition on the sign of the momenta, in particular[9]:

$$p_2 < 0 < p_1. \tag{3.155}$$

From this one immediately deduces that $J(\omega_1, \omega_2) = \frac{1}{2}$, so that the scattering amplitude and the **S** matrix reduce to:

$$\begin{aligned} \mathbf{T}_{ij;kl} &= \frac{1}{8\omega_1\omega_2}M_{ij;kl}(p_1,p_2,p_1,p_2), \\ \mathbf{S}_{ij;kl} &= \delta_{ik}\delta_{jl} + i\mathbf{T}_{ij;kl}. \end{aligned} \tag{3.156}$$

In the specific case of the string in uniform light cone gauge we obtain:

$$\mathbf{S}_{ij;kl} = \{1 + i(1-2a)p_1p_2\}\delta_{ik}\delta_{jl}. \tag{3.157}$$

The physical interpretation of what we have obtained is that, from the point of view of the 1+1 QFT, two particles with momenta p_1 and p_2, by interacting, necessarily undergo an elastic collision of pure

[9] This fact is consistent with the idea that massless particles in a 1d theory always move with the same speed, and can collide only if they have opposite directions.

transition, leaving the interaction again as two particles with momenta p_1 and p_2. The fact that the tensor structure of the Scattering matrix simplifies in the case of the string is no accident. It is a symptom that its quantum theory is integrable, that is, such that the many-particle dynamics can be reduced to the dynamics of two particles. This condition is satisfied if the *Yang-Baxter equation* holds (on this point see for example [8]):

$$\mathbf{S}_{12}\mathbf{S}_{13}\mathbf{S}_{23} = \mathbf{S}_{23}\mathbf{S}_{13}\mathbf{S}_{12}. \tag{3.158}$$

Where $1, 2, 3$ are the indices assigned to three different particles, and the matrix $\mathbf{S}_{ij}$ describes the scattering of the pair (i, j). We observe that, given the simplified structure $\delta_{ik}\delta_{jl}$, the Yang-Baxter equation is trivially satisfied, in support of the integrability claim.

3.3 Observables of the theory

As has been observed, both the volume of the theory R and the Scattering matrix $\mathbf{S}$ (as well as the momentum density P_+) depend on the gauge choice adopted, and therefore cannot be regarded as physical observables. In particular, the dependence of the volume of the theory on the state and on the gauge parameter a is a phenomenon observed in $T\bar{T}$ deformations, as shown in [9] and [10]. What we expect is that, for a finite-volume theory, the spectrum of the Hamiltonian H must be an observable; in particular we expect that:

$$\frac{\mathrm{d}}{\mathrm{d}a}H = 0 = \frac{\mathrm{d}}{\mathrm{d}a}\int_{-R(a)/2}^{+R(a)/2} \mathrm{d}\sigma\, P_+(a). \tag{3.159}$$

Indeed one finds that a change of the parameter a changes the volume precisely in such a way as to compensate the variation of P_+. Let us now show that the spectrum of the Hamiltonian indeed does not depend on a. The expansion of the $\mathbf{S}$ matrix in (3.157) can be seen as an expansion in $T^{-1/2}$. In this way:

$$\mathbf{S}_{ij;kl} = \left(1 + i\frac{1-2a}{T}p_1p_2 + O\left(\frac{1}{T^2}\right)\right)\delta_{ik}\delta_{jl}. \tag{3.160}$$

The subsequent terms can be obtained by carrying out the loop computations[10] over the possible configurations. A possible completion is given by the exponential:

$$\mathbf{S}(p_1, p_2)_{ij;kl} = e^{i\frac{(1-2a)}{T}p_1p_2}\delta_{ik}\delta_{jl}. \tag{3.161}$$

From the dispersion relation (3.108):

$$\mathbf{S}(p_i, p_j) = e^{i(1/2-a)(\omega_i p_j - \omega_j p_i)}\delta_{ik}\delta_{jl}. \tag{3.162}$$

Using the convention that the arguments of the $\mathbf{S}$ matrix are ordered by decreasing magnitude $p_i > p_j$. We observe that the Scattering matrix can be written so that it is manifestly relativistically invariant, by writing it in terms of the Mandelstam variable s:

$$\mathbf{S}_{ij;kl} = e^{-i\frac{(1-2a)}{4T}s}\delta_{ik}\delta_{jl}. \tag{3.163}$$

Let us now consider the gauge choice $a = \frac{1}{2}$, in which the Scattering matrix is equal to the identity. In this case the particles behave as free, non-interacting particles in a box. Given the periodicity, it is necessary to impose the quantisation condition of the momenta:

$$e^{ip_iR} = 1 \rightarrow \quad p_i = \frac{2\pi n_i}{R}, \qquad n_i \in \mathbb{Z}. \tag{3.164}$$

Taking into account the dependence of R on H (2.89):

$$p_i = \frac{2\pi n_i}{J + \frac{1}{2}H}. \tag{3.165}$$

[10] When carrying out the loop computations one observes both infrared divergences (low energies), due to the massless nature of the dispersion relation, and ultraviolet ones (high energies), which require a regularisation.

Now let $H = \sum_j \omega_j$, and calling $N := \sum_j |n_j|$ one obtains the identity:

$$H = \frac{2\pi N}{J + \frac{1}{2}H}, \qquad H = -J + \sqrt{J^2 + 4\pi N}. \tag{3.166}$$

This condition can be refined by considering the *level matching condition* (3.167):

$$p_{ws} = \sum_j p_j = 0. \tag{3.167}$$

Now let

$$N_> := \sum_{j|\, n_j>0} n_j, \qquad N_< := \sum_{j|\, n_j<0} n_j, \tag{3.168}$$

one obtains the explicit expression of the Hamiltonian spectrum:

$$H = -J + \sqrt{J^2 + 8\pi N_>}. \tag{3.169}$$

In the case of generic a, one can show that the boundary condition (3.164) generalises through the *Bethe-Yang equation* [9]:

$$e^{ip_i R} \prod_j \mathbf{S}(p_i, p_j) = 1. \tag{3.170}$$

The relation above can be justified by thinking that, by making the i-th particle go once around the circle, it interacts with the other particles, and the result of each interaction is a factor $\mathbf{S}(p_i, p_j)$. Making the various terms explicit we obtain:

$$\begin{aligned} &e^{ip_i(J+(1-a)H)} \prod_j \mathbf{S}(p_i, p_j) = \\ &e^{ip_i(J+\frac{1}{2}H)} e^{ip_i(\frac{1}{2}-a)H} e^{i(\frac{1}{2}-a)(\sum_j \omega_i p_j - \sum_j \omega_j p_i)} = \\ &e^{ip_i(J+\frac{1}{2}H)} e^{ip_i(\frac{1}{2}-a)H} e^{i(\frac{1}{2}-a)(-Hp_i)} = \\ &e^{ip_i(J+\frac{1}{2}H)} = 1. \end{aligned} \tag{3.171}$$

Where from the second to the third step we used $\sum_j \omega_j = H$ and the level matching condition $\sum_j p_j = 0$. We see that the result obtained does not depend on a and coincides with the boundary condition studied previously, which confirms that the spectrum (3.169) is valid for every value of a. Finally, we observe that this treatment does not take into account finite-volume effects, which correct the energy of the state. In particular, these effects come from the *zero point energy*, that is, from the choice of neglecting the constant terms that would appear in the Hamiltonian because of the commutation relations of the a and $a^\dagger$. To make sense, these contributions require a regularisation, and they add a negative term to the spectrum of the Hamiltonian:

$$H = -J + \sqrt{J^2 + 4\pi N} - .... \tag{3.172}$$

Nevertheless, one can show [10] that if a fermionic contribution is added to the theory studied (also called the bosonic theory), the spectrum of the Hamiltonian undergoes a new variation that cancels the previous one, making the expression (3.169) exact.

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